Upcoming + Reminders

Find all links on the **course website:** cogs9.github.io/fa26

📝 **Assignments** (find links on course website):

- Reading Quiz 4 (DUE: WEDNESDAY)

All due at 11:59pm

- Assignment 1— Data Visualization (DUE: Wed week 7)
- Assignment 2 — p-hacking (DUE: Wed week 8)
- Reading Quiz 5 (DUE: Wed week 9)
- Assignment 3 — Machine learning (DUE: Wed week 10)

Midterm 2 - Monday Week 9 (November 23)

COGS 9 | Week 6

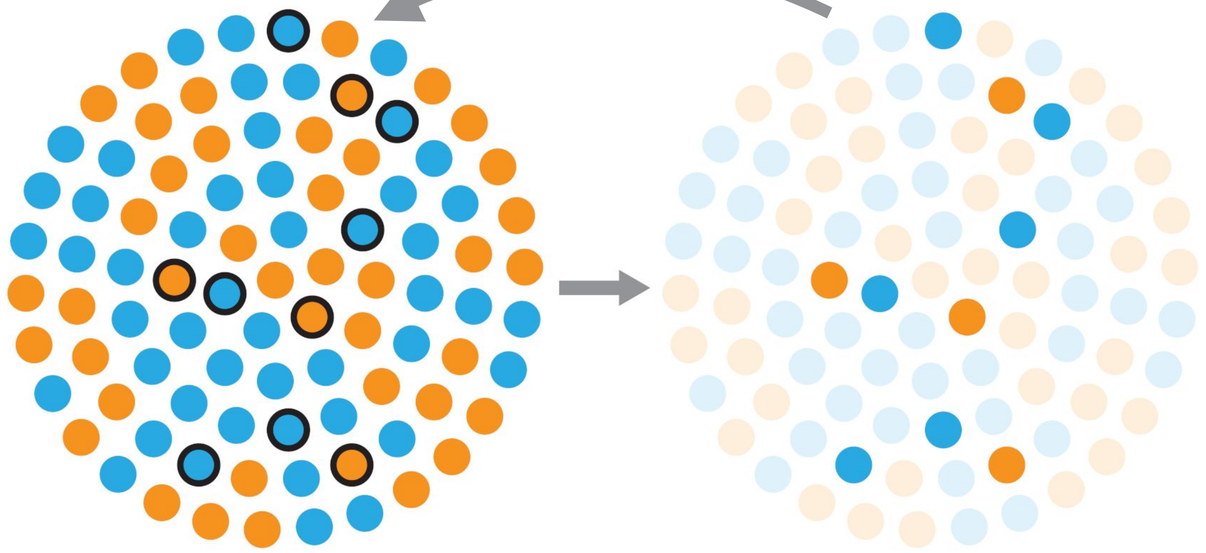
Machine Learning: Supervised Learning

Brendan Tomoschuk

Last week: Statistical Inference

- Statistical inference is the process of drawing conclusions about a **population** from a **sample** of data
- Statistical inference uses *sample statistics* to estimate *population parameters*

Inference



Population

A parameter is a characteristic of the population

(e.g., the *population* mean, std, variance)

Sample

A statistic is a characteristic of the sample

(e.g., the *sample* mean, std, variance)

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course code:

WSTAPX

Which statistical test is most appropriate for assessing whether two categorical variables are associated?

(A) t-test

(B) ANOVA

(C) Pearson correlation

(D) Chi-squared

(E) I’m not sure / confused

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course code:

WSTAPX

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course code:
WSTAPX

A study reports a 95% confidence interval (CI) for the mean height of COGS 9 students as [67, 69] inches. Which statement is TRUE?

(**A**) 95% of students' heights fall between 67 and 69 inches

(**B**) Out of 100 samples, 95 of them would have 95% CIs that contain the true mean population height

(**C**) There is a 95% probability that the true mean population height is between 67 and 69 inches

(**D**) Out of 100 samples, 95 of them would have a mean height between 67-69 inches

(**E**) I'm not sure / confused

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course code:
WSTAPX

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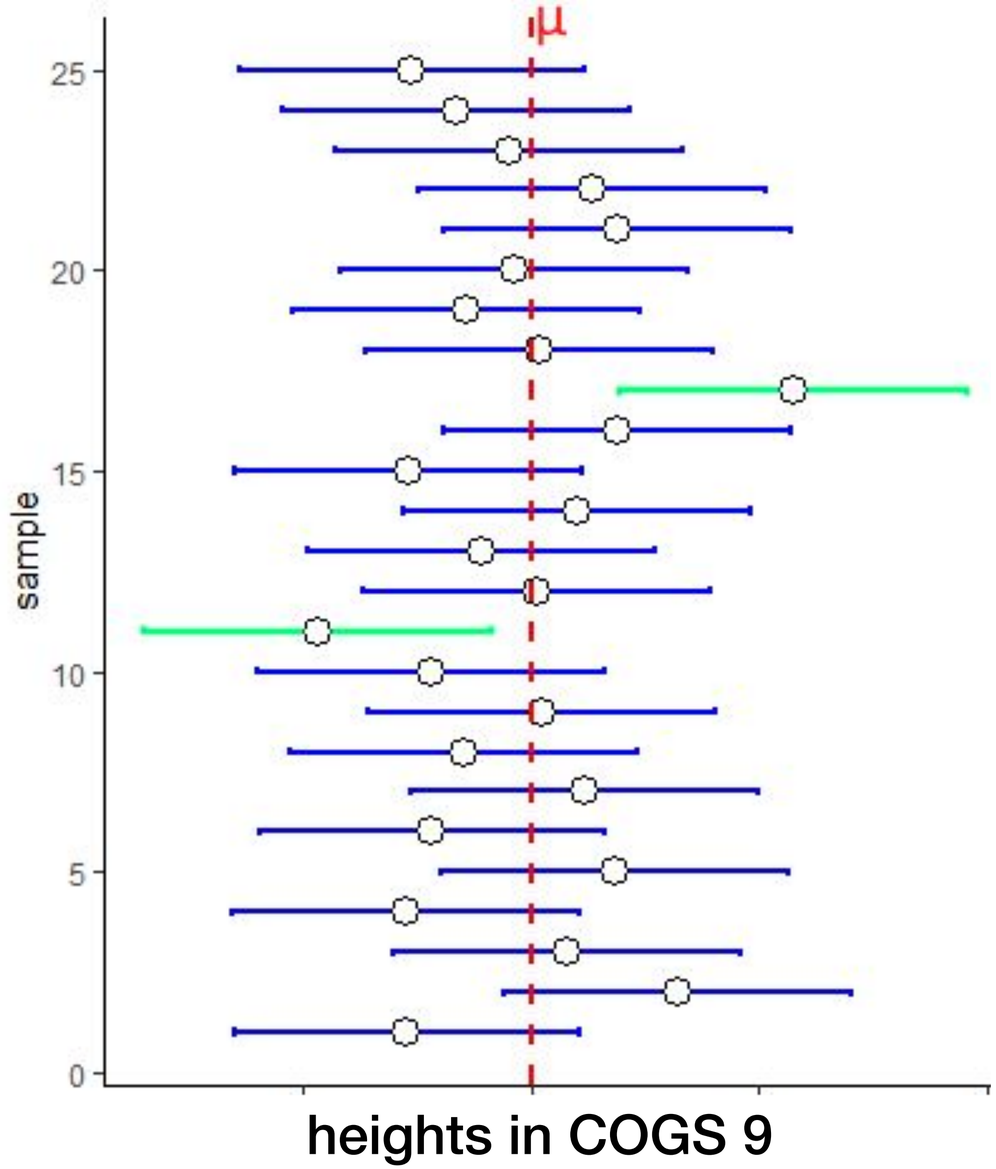
(**E**) I'm not sure / confused

Quantifying uncertainty of a sample statistic

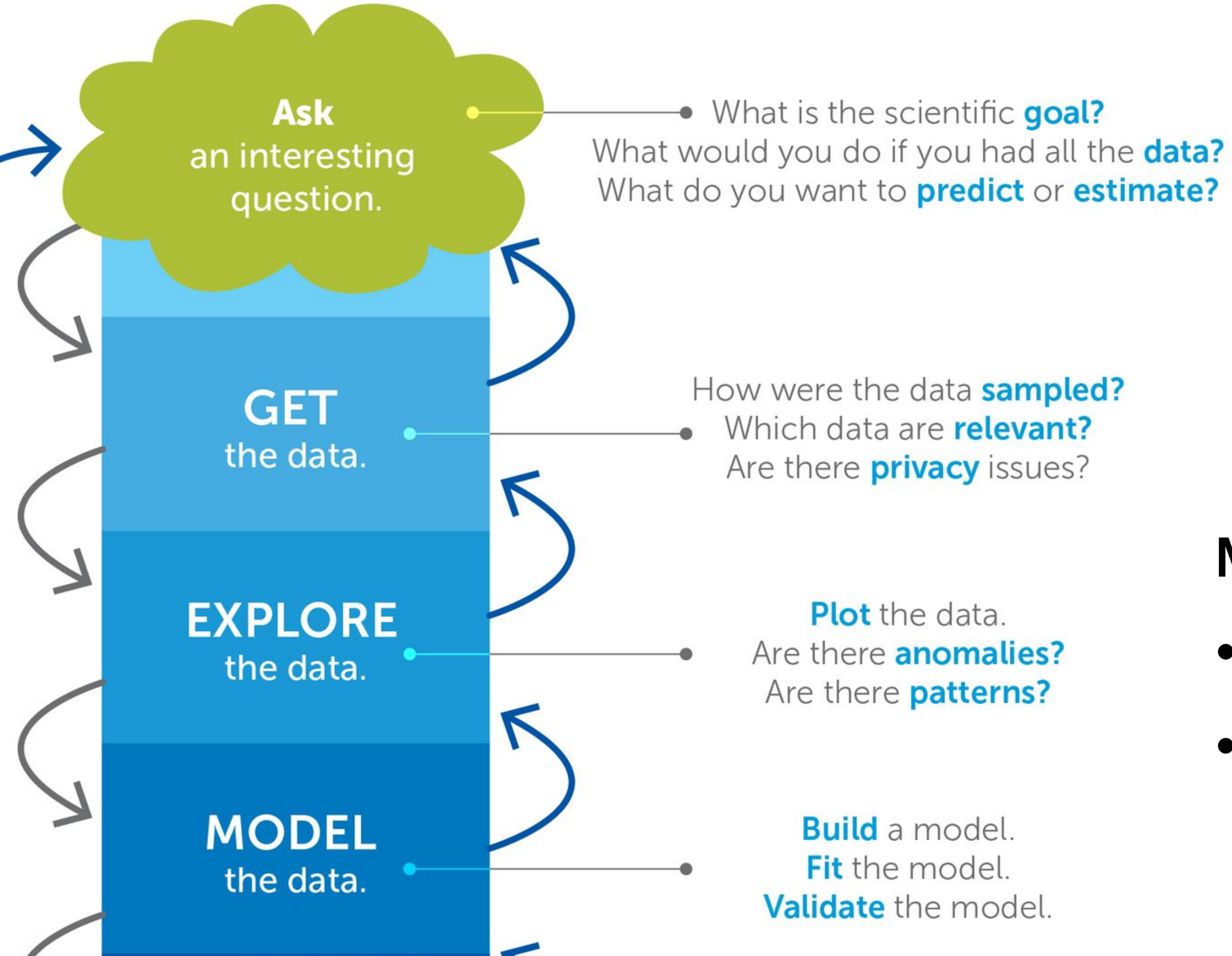
A 95% confidence interval (CI) means that if you…

1. Take **100 samples** from the population
2. Calculate a 95% CI for each sample:
   $[\text{mean} \pm 1.96 \times \text{SE(mean)}]$

- 95 out of those 100 CIs would contain the true population mean height
- 5 out of those 100 CIs would NOT contain the true population mean height



Today: Machine Learning



Modeling the data in order to

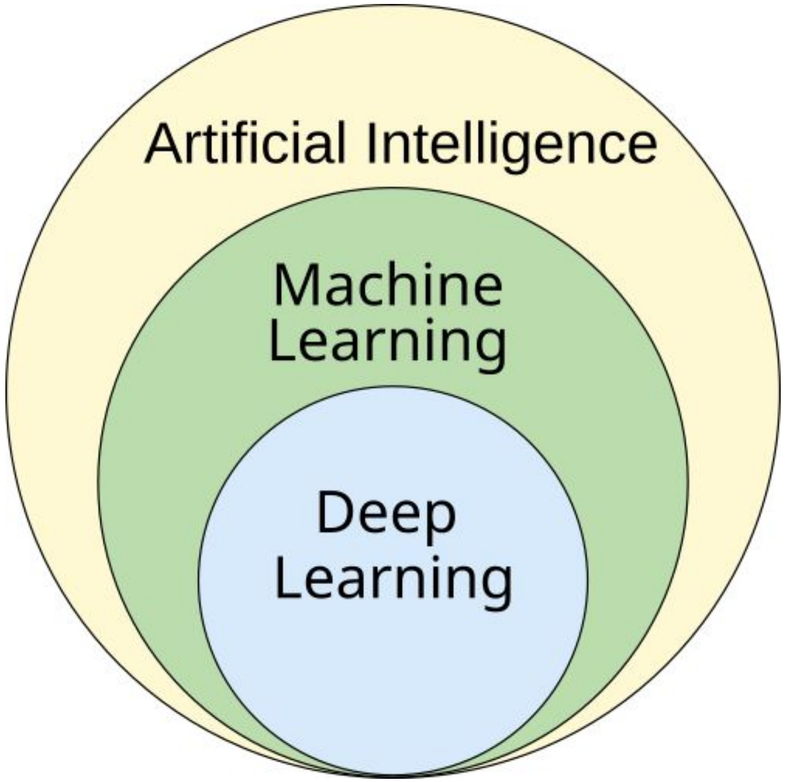
- make predictions on new data
- make decisions based on those predictions

Learning Goals

1. Define the goal of machine learning and distinguish its major paradigms: *supervised, unsupervised, and reinforcement learning*
2. Differentiate supervised learning problem types (*regression vs. classification*)
3. Describe when to apply each of the following models:
    - *simple & multiple linear regression*
    - *non-linear regression*
    - *logistic regression*
    - *decision trees*
    - *neural networks*

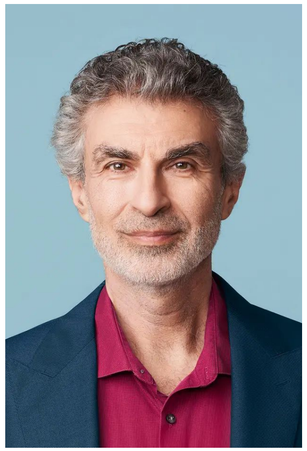
Artificial intelligence and machine learning

- **Artificial intelligence (AI)** is field that is concerned with enabling machines to simulate human intelligence
- **Machine learning (ML)** is subfield of AI that develops models to **learn patterns from data** to make ***predictions*** or ***decisions***
- **Deep learning (DL)** focuses on using multi-layered (“deep”) neural networks from large amounts of data



Pioneers of the AI revolution

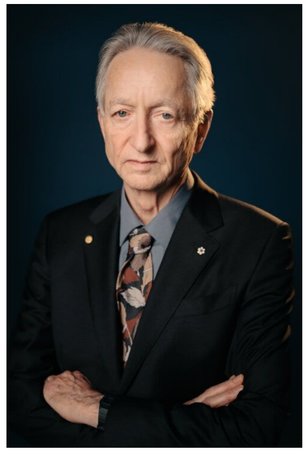
2024 Nobel Prize in Physics



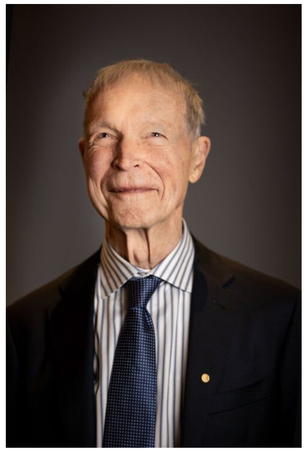
Yoshua Bengio



Yann LeCun



Geoff Hinton



John Hopfield

2019 ACM Turing Award

“Godfathers of AI”

Created ImageNet



Fei-Fei Li

MacArthur Fellow



Daphne Koller



Raia Hadsell



Machine Learning

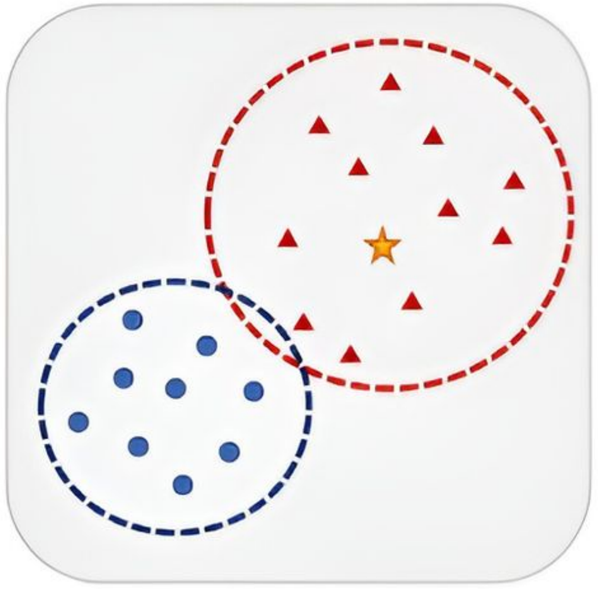
Learning from data to predict or generalize to unseen data

Learning from trial-and-error interactions with the environment

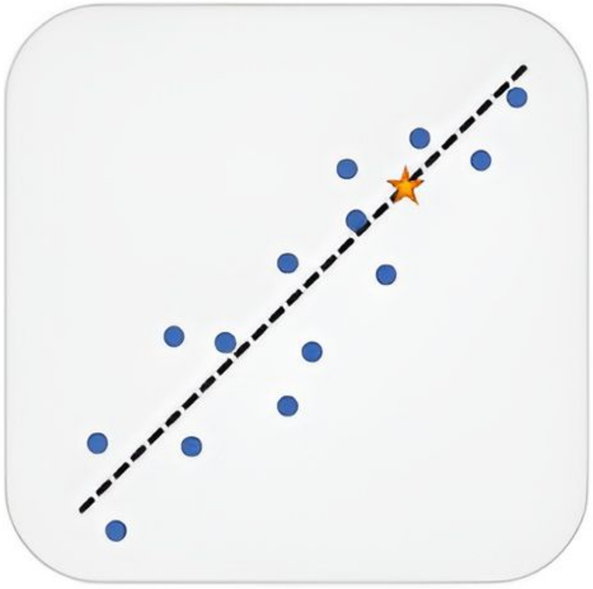
Supervised Learning

Model training with labelled data

Classification



Regression

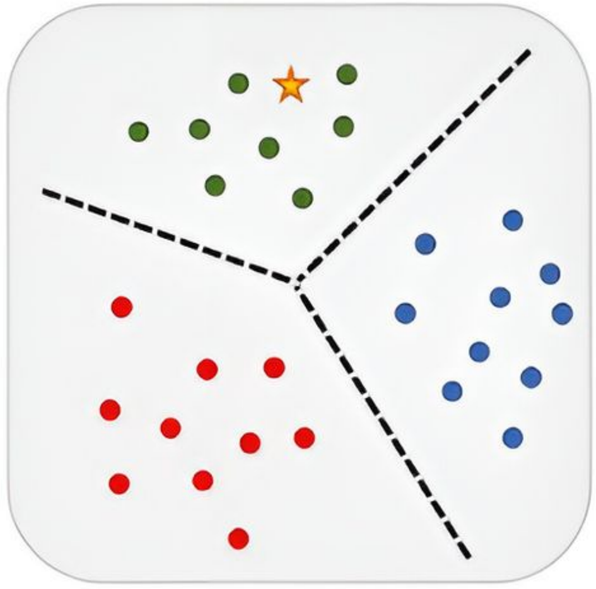


Unsupervised Learning

Model training with unlabelled data

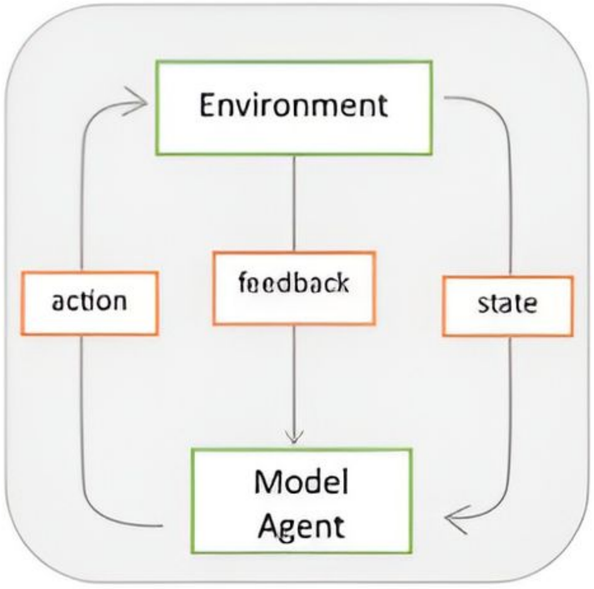
Discovering patterns in data

Clustering



Reinforcement Learning

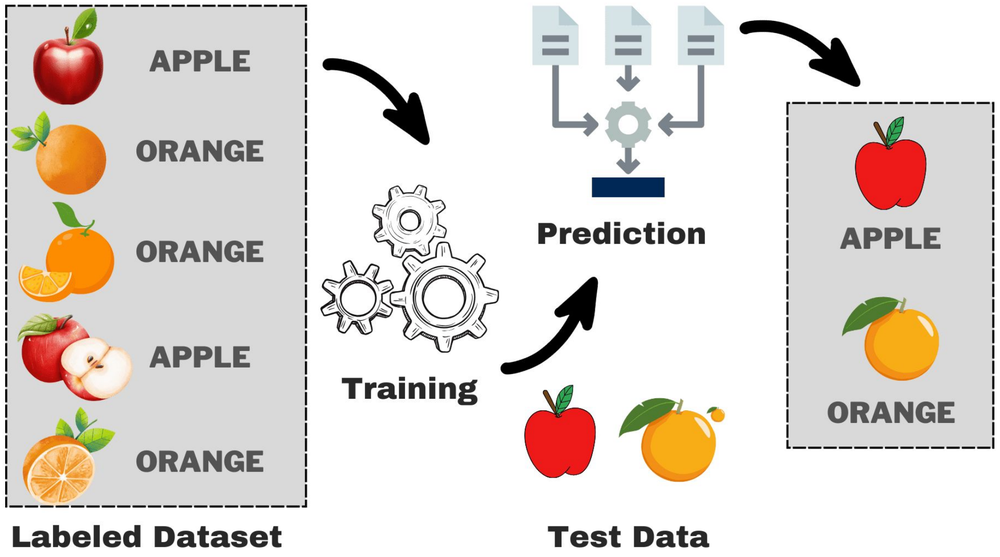
Model take actions in the environment then received state updates and feedbacks



Supervised vs. Unsupervised Learning

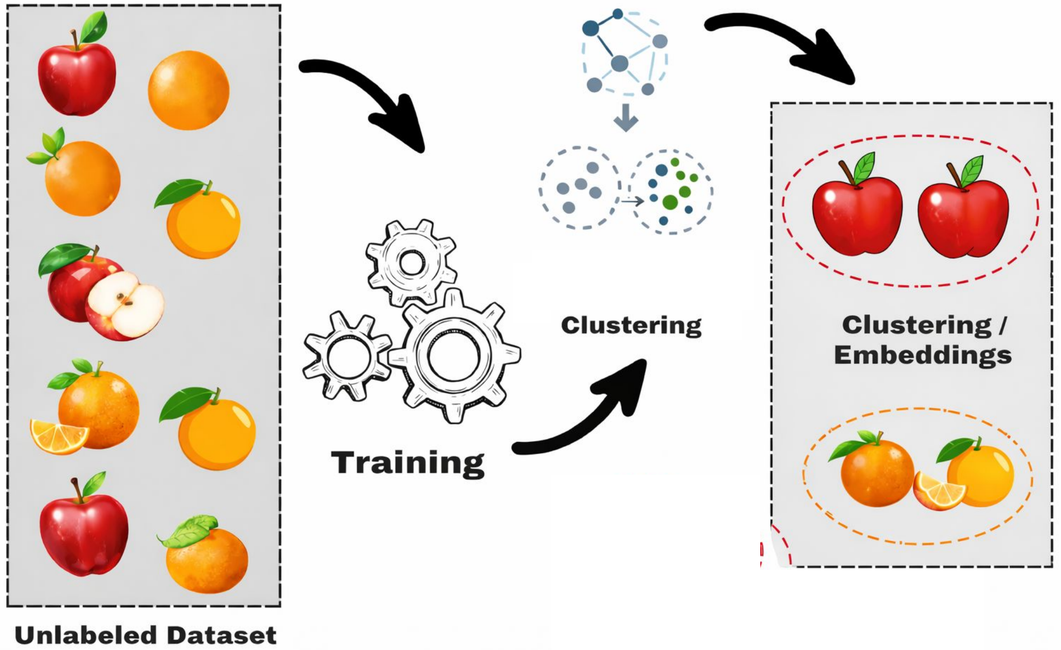
Supervised: there is a correct answer or label to be predicted

- **Goal:** correctly predict labels for each data point
- Two main types: regression & classification



Unsupervised: no labels or correct answers to predict

- **Goal:** find patterns and structure in unlabeled data by **grouping** or **clustering** data by similarity



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course code:
WSTAPX

Which of the following is a supervised learning problem?

(**A**) Predicting house prices from square footage and location

(B) Grouping people into personality types based on survey responses

(C) Predicting whether someone will pass or fail COGS 9

(D) All of the above

(E) A and C only

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Regression vs. Classification (Supervised)

Regression: predicting quantitative responses

Classification: predicting qualitative (categorical) responses

unknown function we are trying to estimate

$$Y = f(X)$$

output, response dependent variable

predictors, features, independent variable

probability (that Y is in category k, given X)

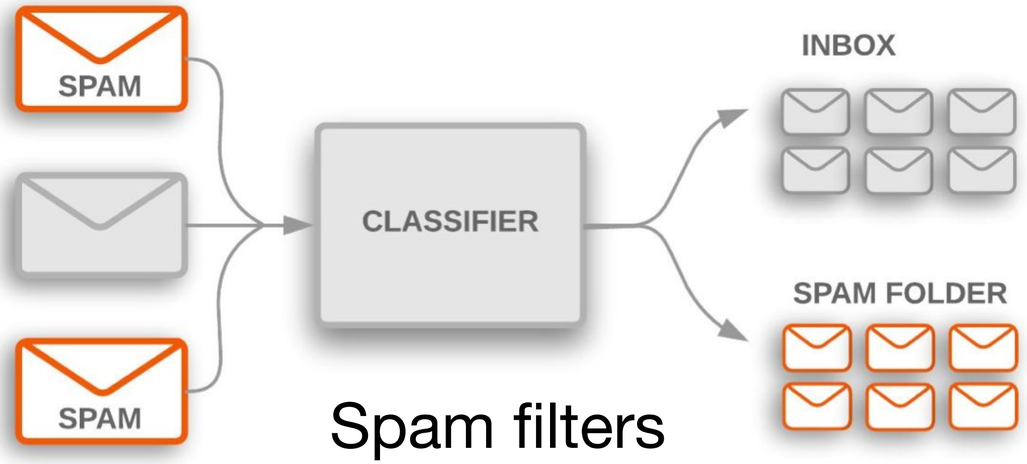
$$P(Y = k \mid X) = f(X)$$

output, response

category or class

Regression vs. Classification (Supervised)

Real-life examples of classification



Spam filters



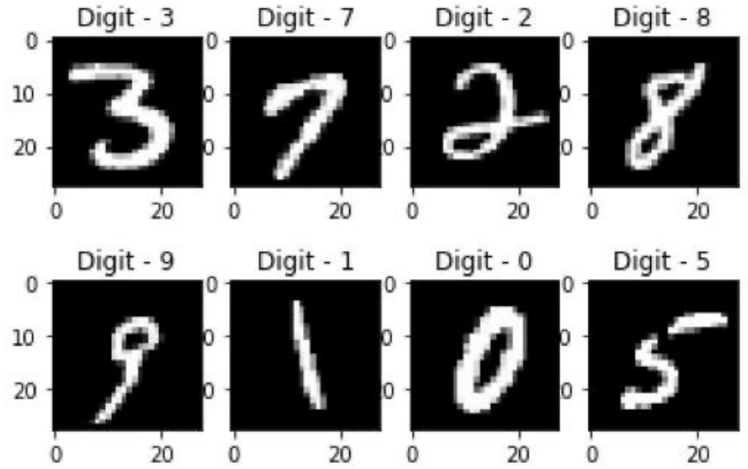
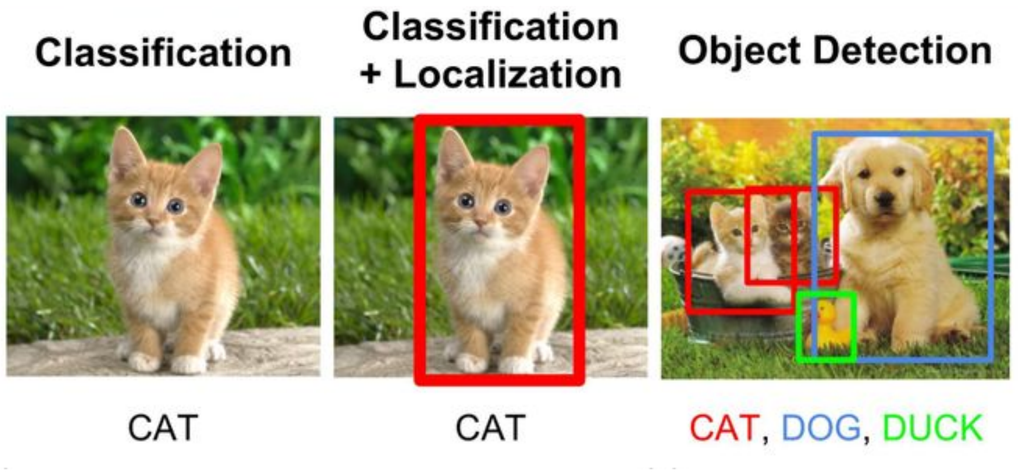
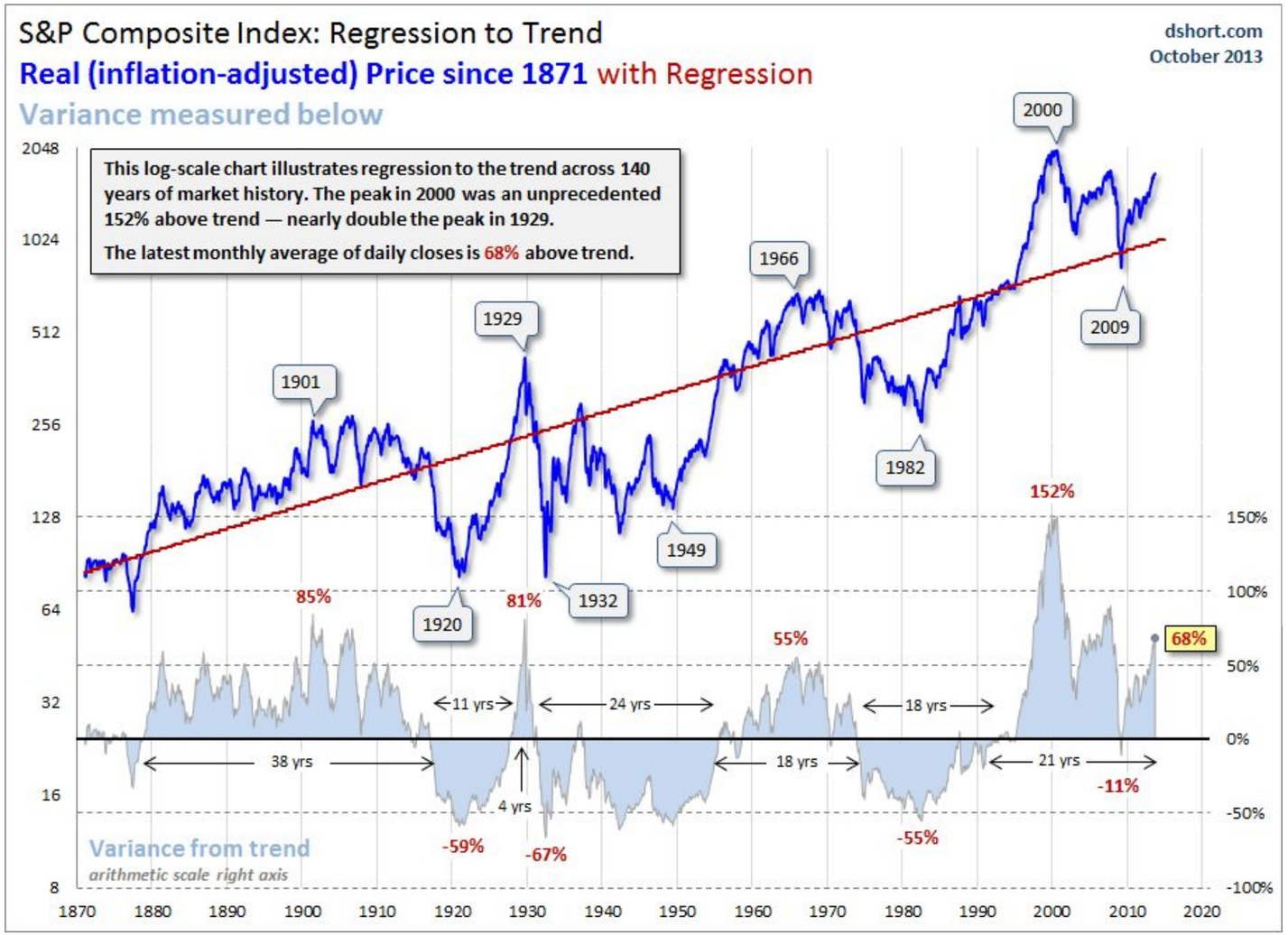


Image & object classification

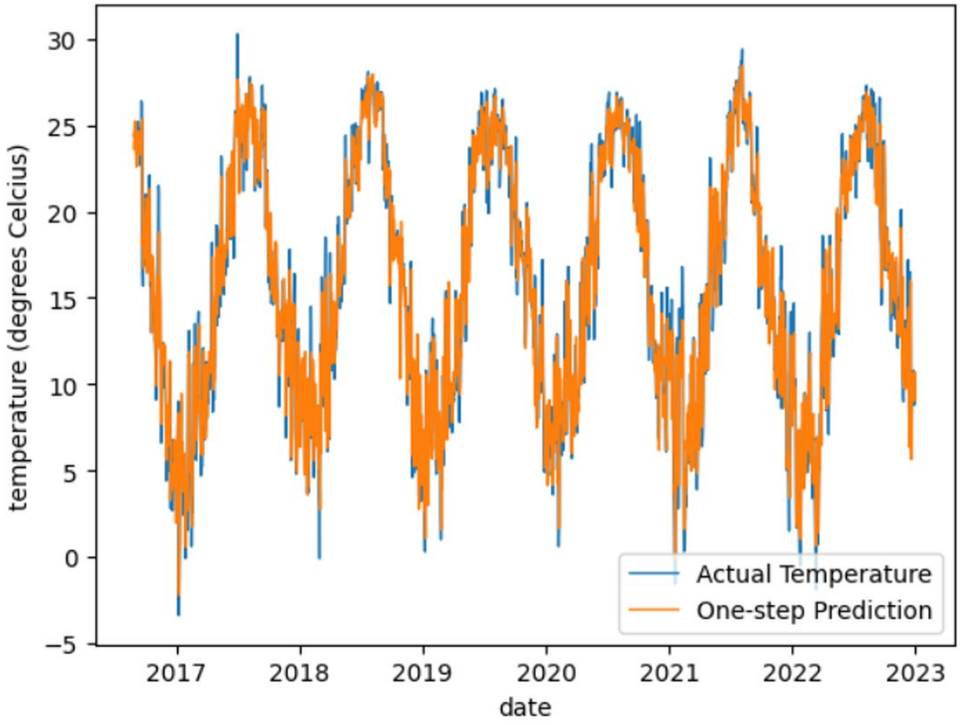


Regression vs. Classification (Supervised)

Real-life examples of regression



Istanbul's Weather Forecast Using LSTM model



Weather prediction

Stock prediction

Regression vs. Classification (Supervised)

Regression: predicting quantitative responses

Examples: predicting…

- height (Y) from weight, gender, etc. (X)
- income (Y) from education level (X)
- neural activity (Y) from motor movements (X)

Classification: predicting qualitative (categorical) responses

Example: predicting…

- whether someone will pass or fail (Y) from their grades, attendance, etc. (X)
- cancer type (Y) from a patient's genes and demographics (X)
- image labels (Y) from RGB values (X)

Regression vs. Classification (Supervised)

Regression: predicting quantitative responses

Models:

- Linear regression (simple & multiple)
- Non-linear regression
- Ridge & lasso regression

Both regression & classification:

- Decision Trees
- Random Forests & Boosting
- K-Nearest Neighbors
- Neural Networks

Classification: predicting qualitative (categorical) responses

Models:

- Logistic regression
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)
- Naive Bayes
- Support Vector Machine (SVM)

Regression vs. Classification (Supervised)

Regression: predicting quantitative responses

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The ones we will discuss!

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course code:
WSTAPX

Which of these would be best suited for a classification model?

(A) Predicting tomorrow's temperature

(B) Predicting whether it will rain or not

(C) Predicting your average commute time to campus

(D) Predicting your exam grade (0-100)

(E) I'm not sure / I'm confused

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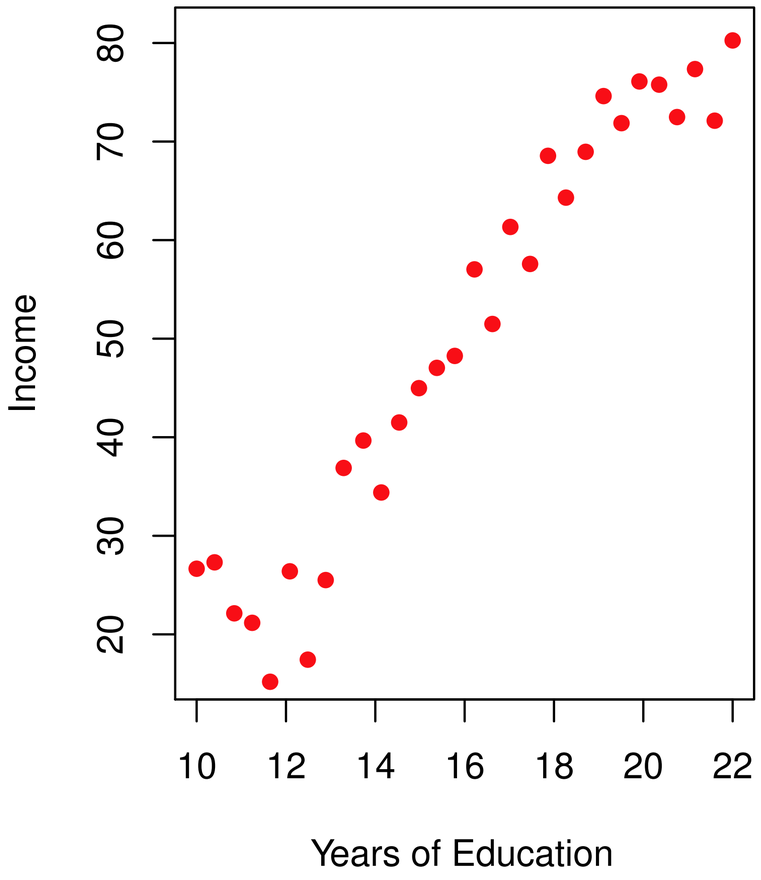
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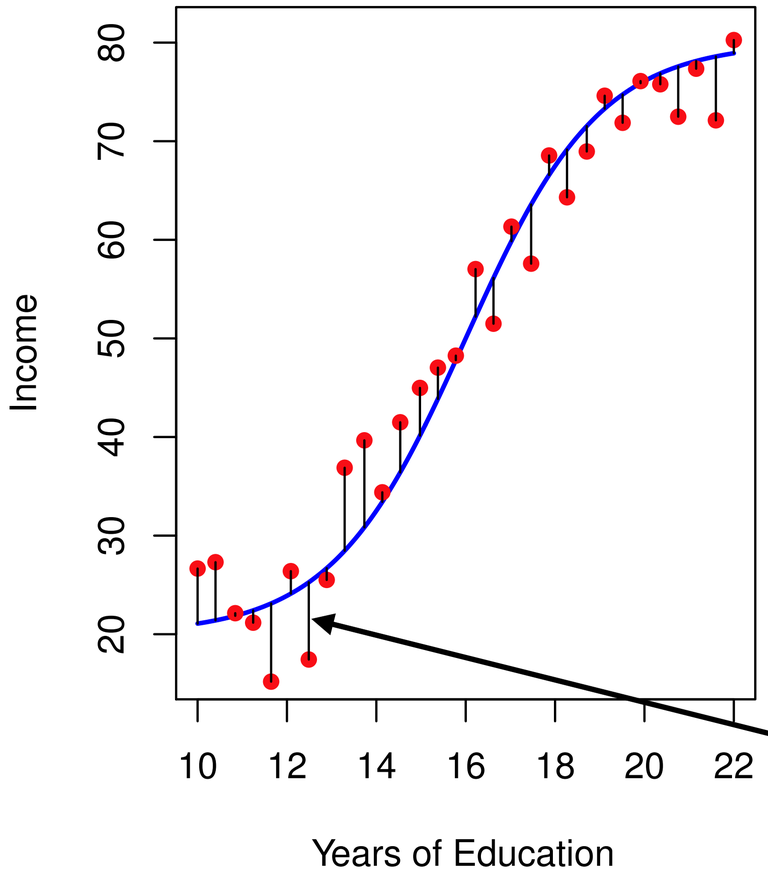
Regression

Predicting quantitative Y

Regression (Predicting quantitative Y)

$$Y = f(X)$$





f : our best guess of the true relationship between income and years of education (estimated from the data)

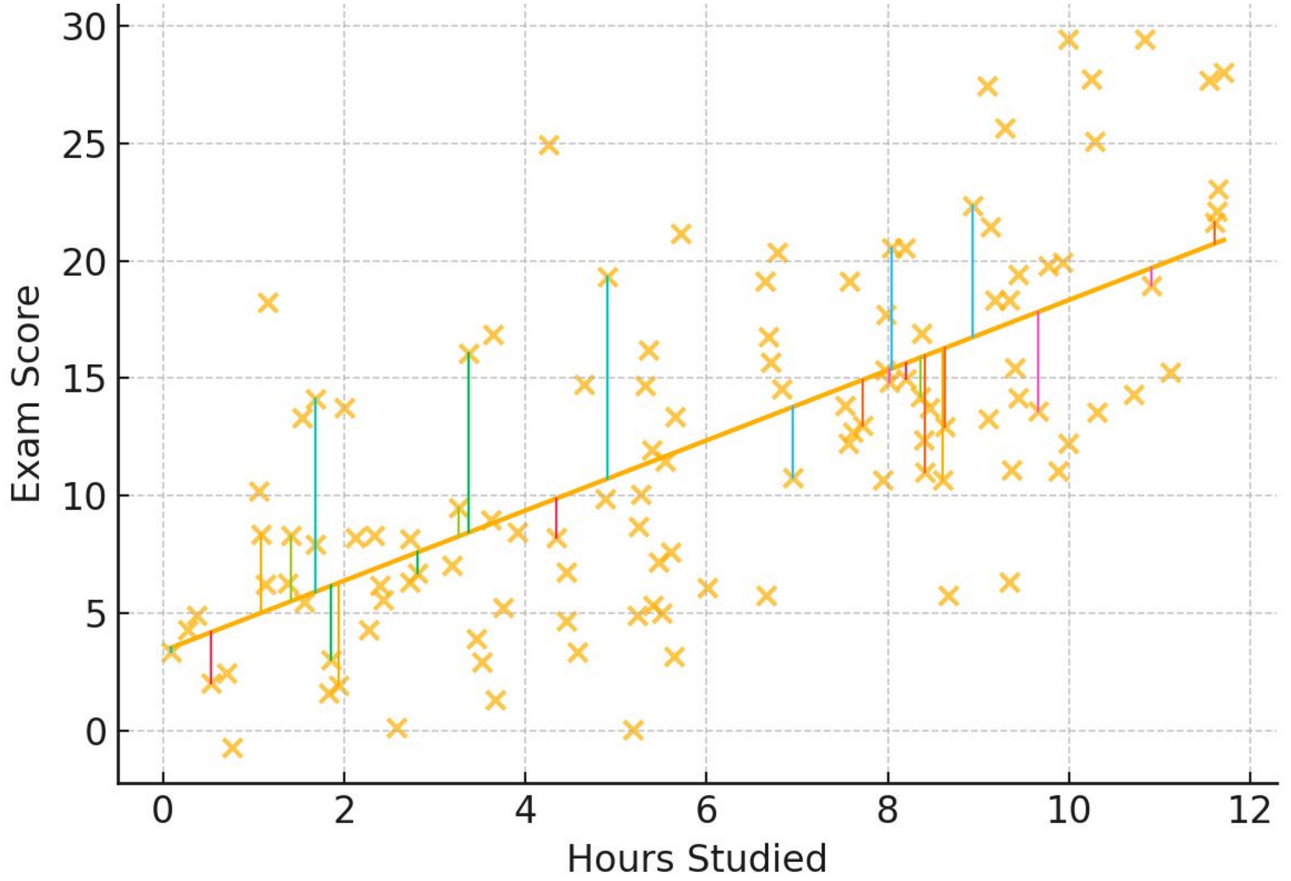
f has a specific form depending on the model

Supervised learning = estimating what f is!

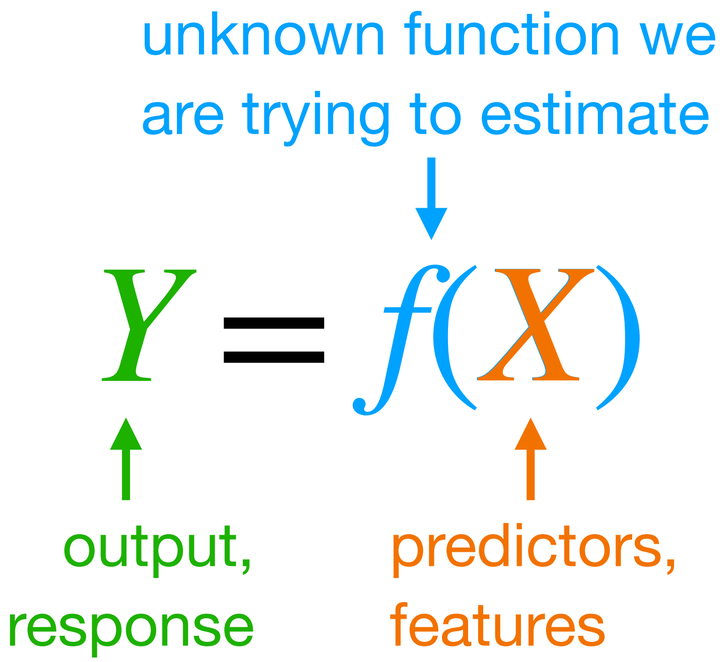
Simple Linear Regression

Simple linear regression: models the relationship between two quantitative variables

- f is a straight line: assumes that relationship between two variables is ***linear***!



Simple Linear Regression



Simple Linear Regression

$$y = mx + b$$

unknown *parameters* we are trying to estimate

$$Y = \beta_0 + \beta_1 X$$

output, response

predictors, features

β_1: "slope" or expected change in Y for a 1-unit increase in X

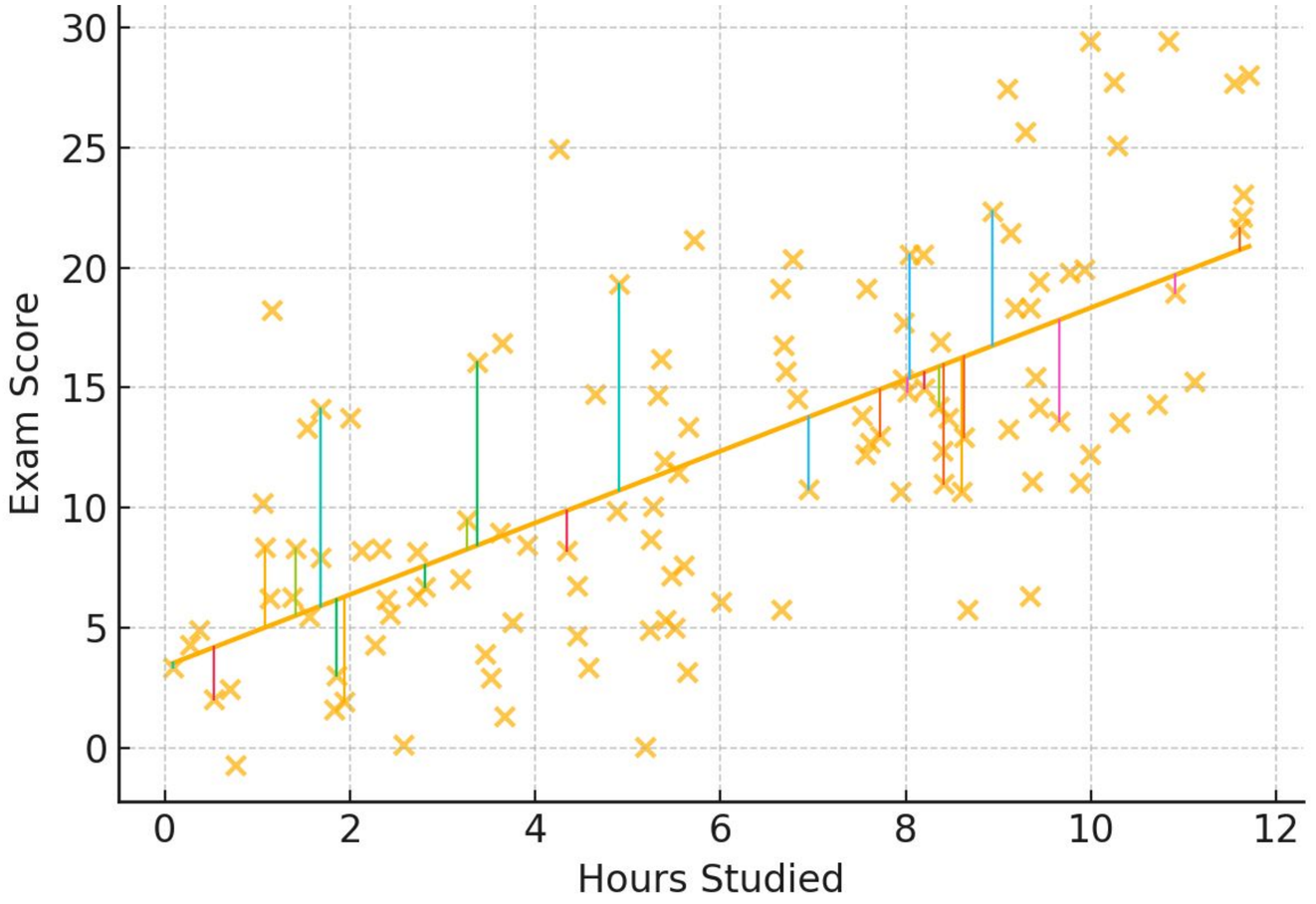
β_0: "y-intercept," or expected Y when $X = 0$

Simple Linear Regression

Choose $\hat{\beta}_0, \hat{\beta}_1$ that minimizes the **errors**, or the **residual sum of squares (RSS)**

$$\text{RSS} = \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2$$

Ordinary least squares (OLS) method



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course code:
WSTAPX

You fit a simple linear regression model:

ExamScore = β_0 + β_1HoursStudied

And find that $\beta_1 = 3.5$. What is the correct interpretation of β_1?

(**A**) The number of hours you need to study to increase your exam score by 1 point

(**B**) The expected exam score when you did not study at all

(**C**) The expected change in exam score for every additional hour studied

(**D**) I'm not sure / I'm confused

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WSTAPX

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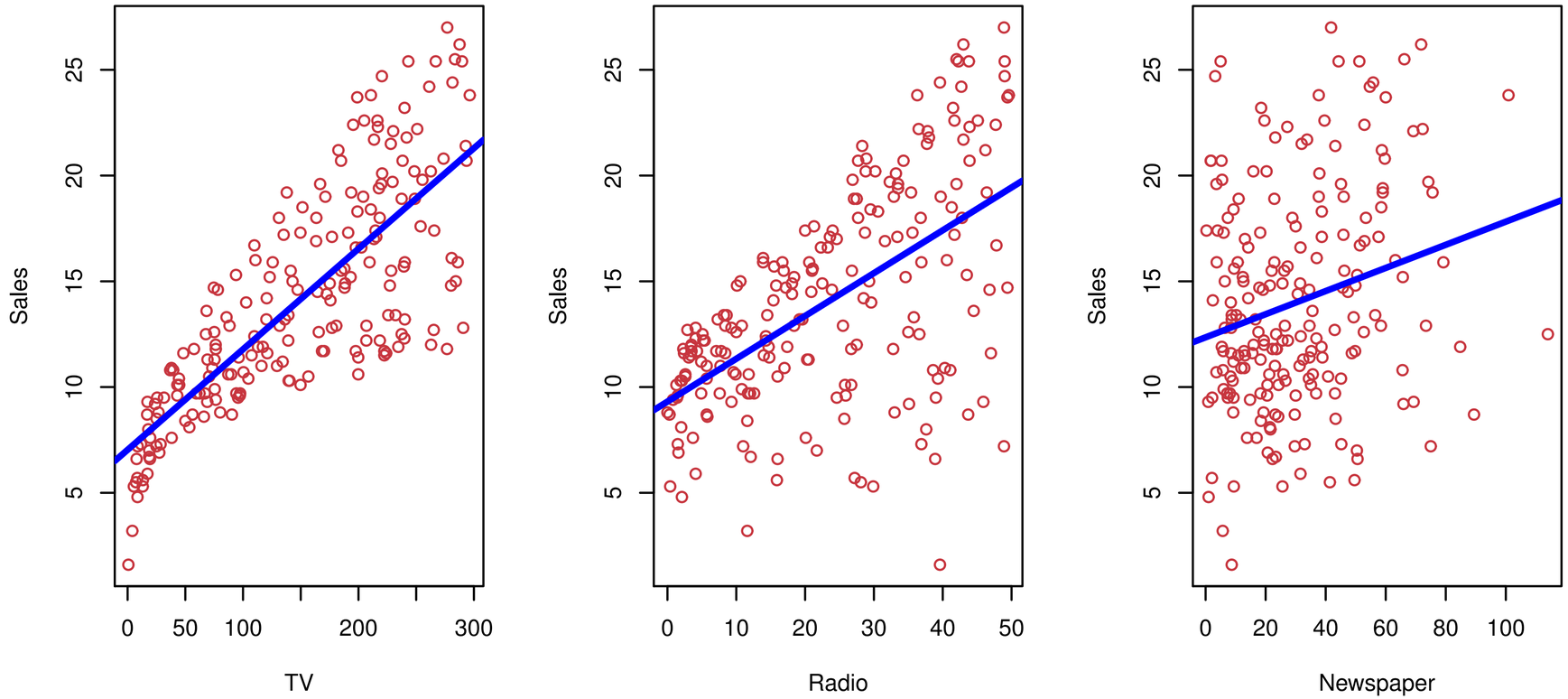
(C) The expected change in exam score for every additional hour studied

(D) I'm not sure / I'm confused

Multiple Linear Regression

What happens when you have many predictors (more than one X)?

- You could run three separate simple linear regressions…but potential issues:
  1. How to best predict Y if you have more than one model?
  2. Each model ignores other two potential predictors



Multiple linear regression: models the relationship between **many** predictors and **one** outcome

Multiple Linear Regression

unknown *parameters* we are trying to estimate

β_j quantifies the association between the predictor X_j and Y

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \epsilon$$

output, response

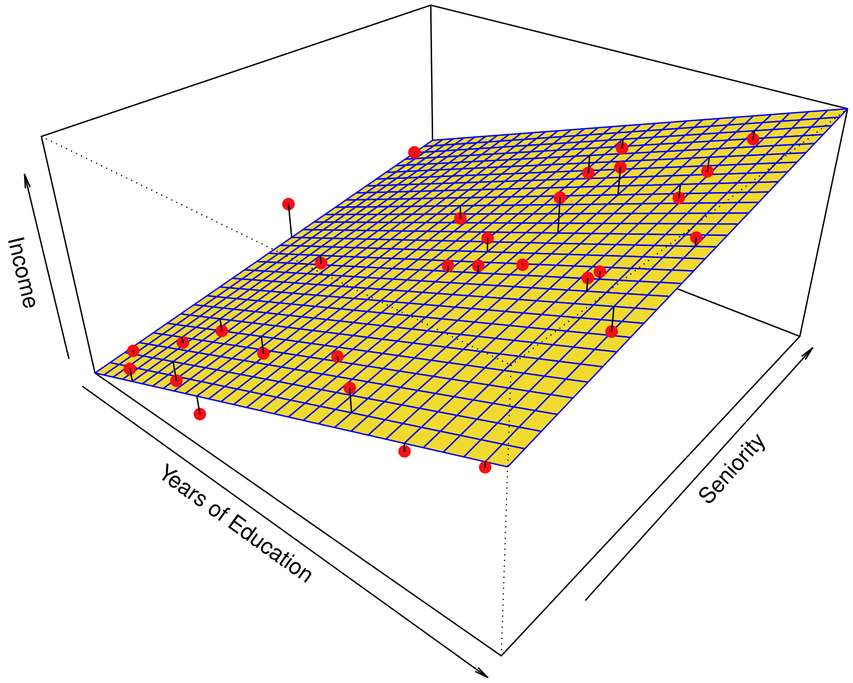
predictors, features

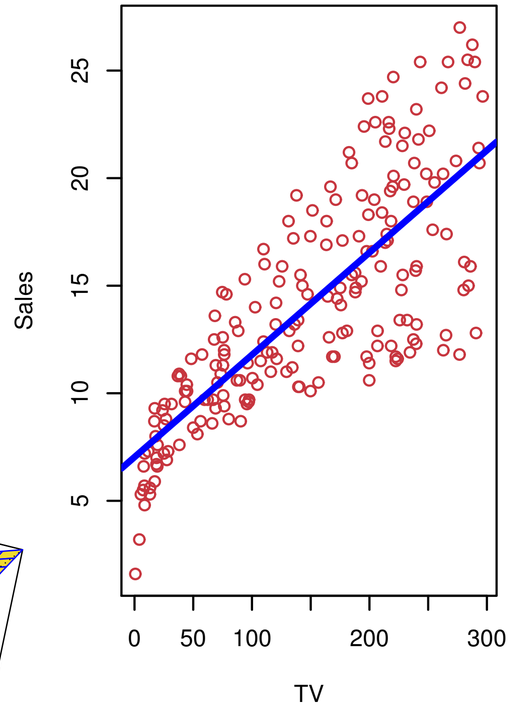
irreducible error

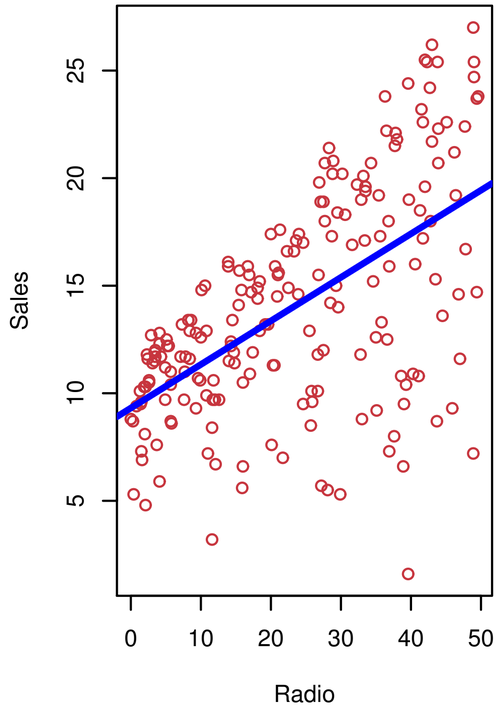
β_j: expected change in Y for a 1-unit increase in X_j, holding other predictors fixed

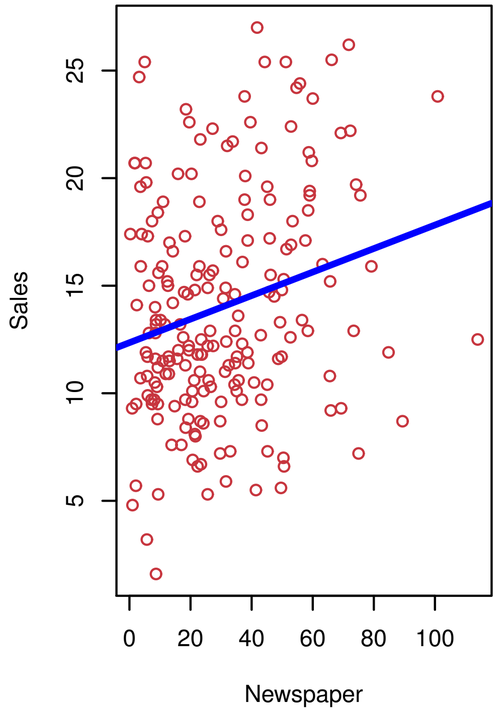
β_0: "y-intercept," or expected Y when $X_1, \ldots, X_p = 0$

MLR Examples







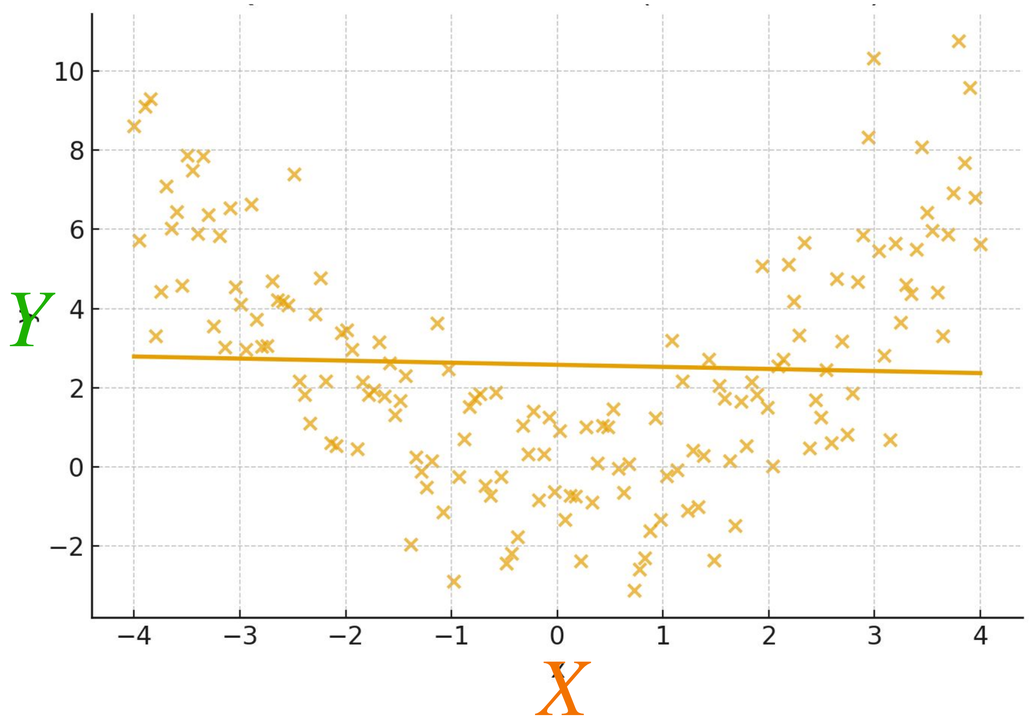


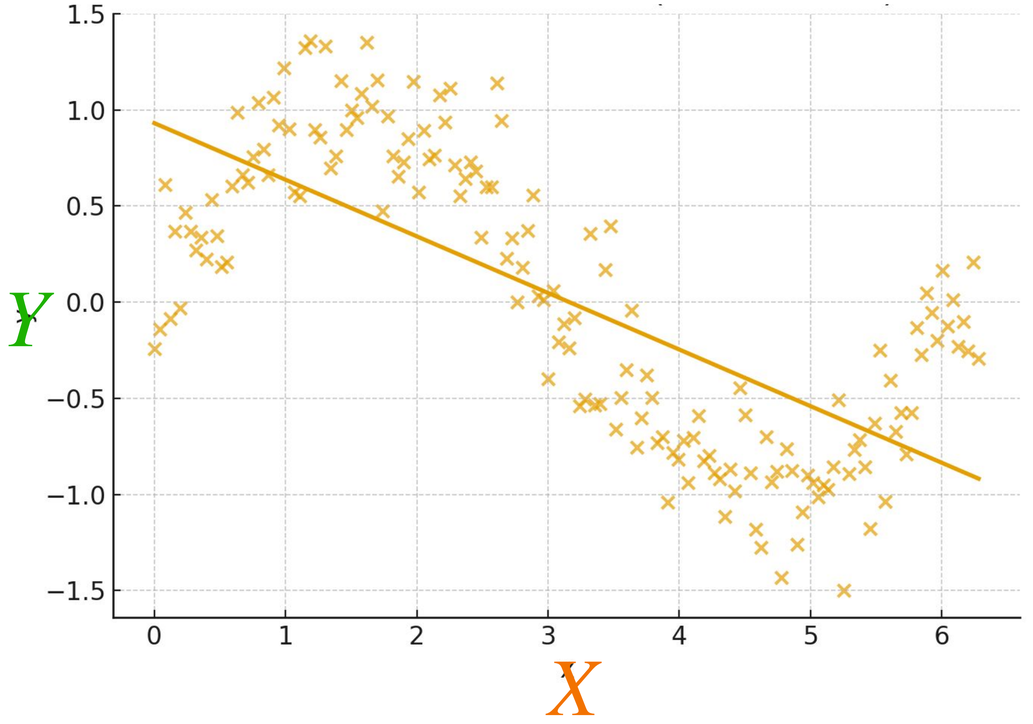
$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper}$$

$$\text{income} = \beta_0 + \beta_1 \times \text{education} + \beta_2 \times \text{seniority}$$

Non-Linear Regression

Relaxing the linear assumption: Linear models are awesome! But sometimes, it's just not going to be enough…

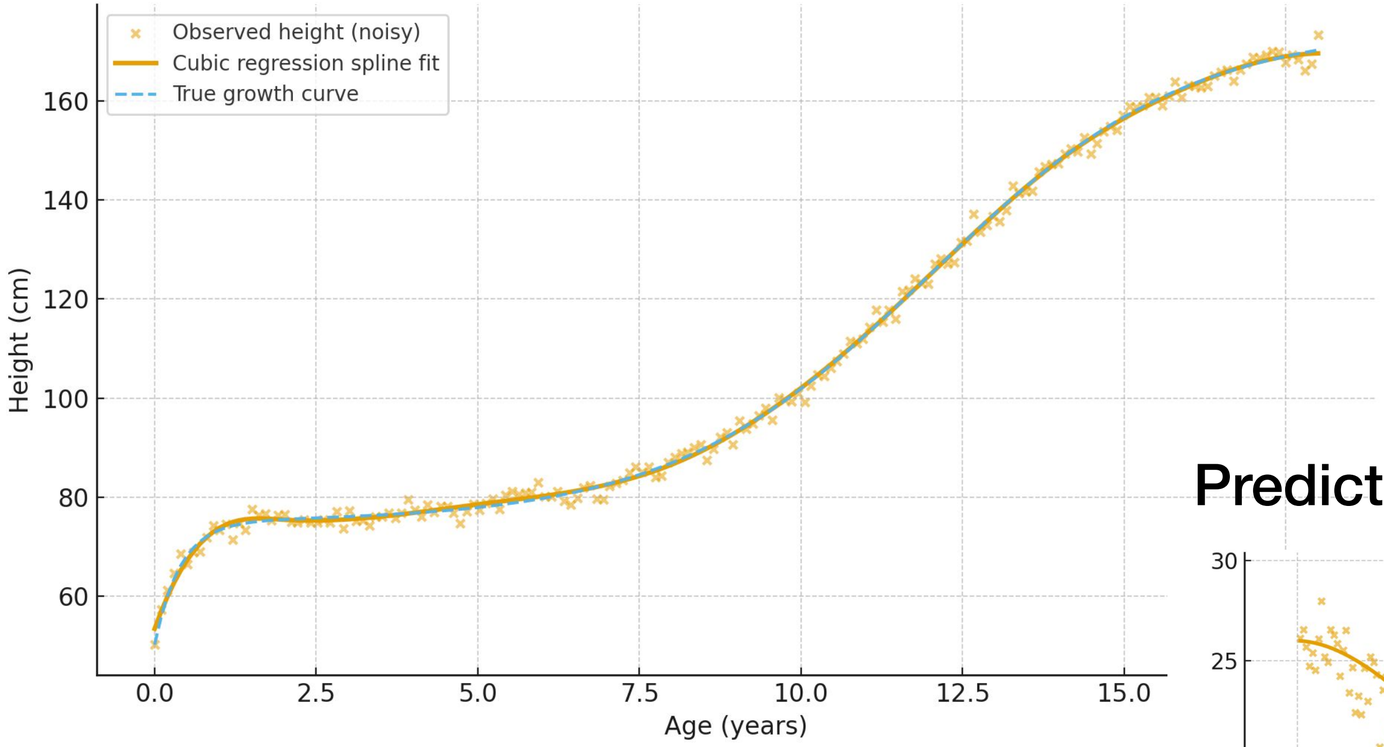




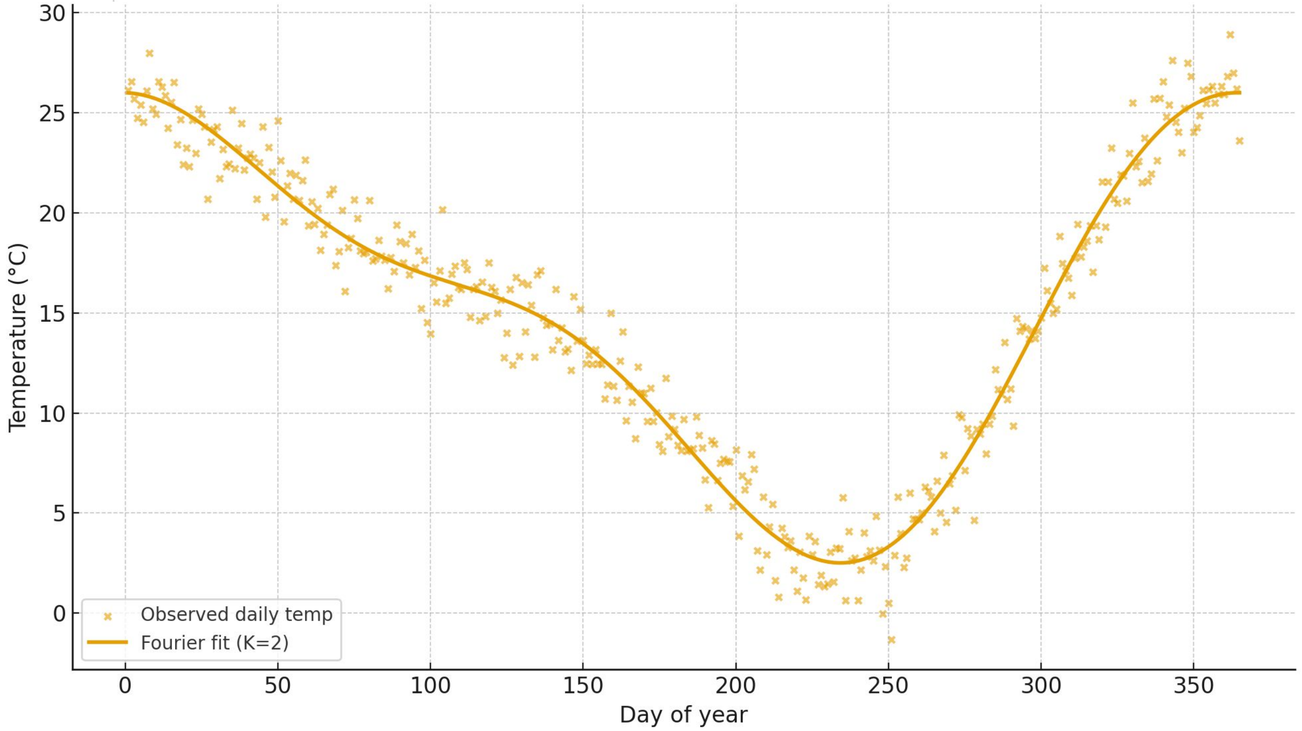
You ***CANNOT*** model these relationships between X and Y with a straight line!

Non-Linear Examples

Predicting a child's height from their age



Predicting temperature from the day of the year



Polynomial Regression

- **Non-linear regression**: any model that assumes that the relationship between Y and X is ***non-linear***!
- **Polynomial regression**: a non-linear regression model that models the outcome Y as a non-linear, polynomial function of X by raising it to a power (e.g., X^2)

Polynomial regression

unknown *parameters* we are trying to estimate

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \ldots + \beta_d X^d$$

output, response

predictors, features

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course code:
WSTAPX

You want to predict a student's GPA from their class standing, hours of sleep, and exam scores. You suspect that the relationship between the variables is linear.

Which kind of model would you use?

(**A**) Simple linear regression

(**B**) Multiple linear regression

(**C**) Polynomial regression

(**D**) I'm not sure / I'm confused

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(B) Multiple linear regression

(C) Polynomial regression

(D) I'm not sure / I'm confused

Classification

Predicting qualitative Y

Classification (Predicting qualitative Y)

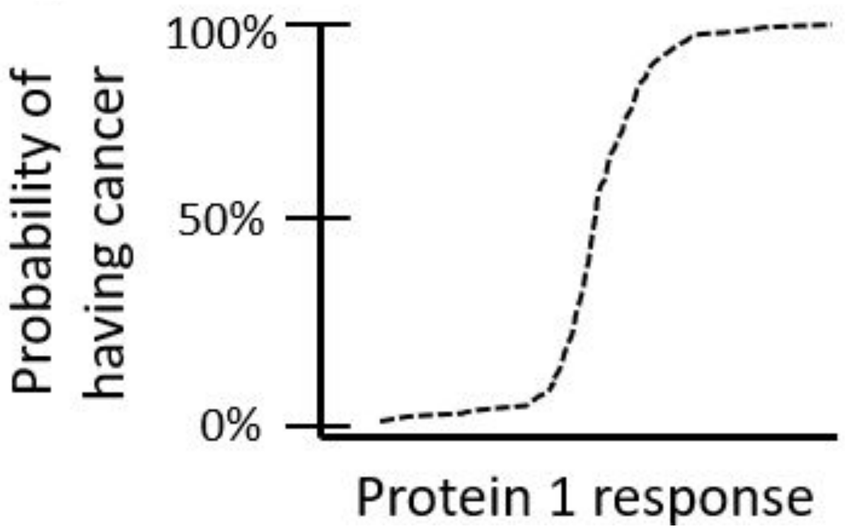
Classification models predict the **probability** of a data point belonging to a category

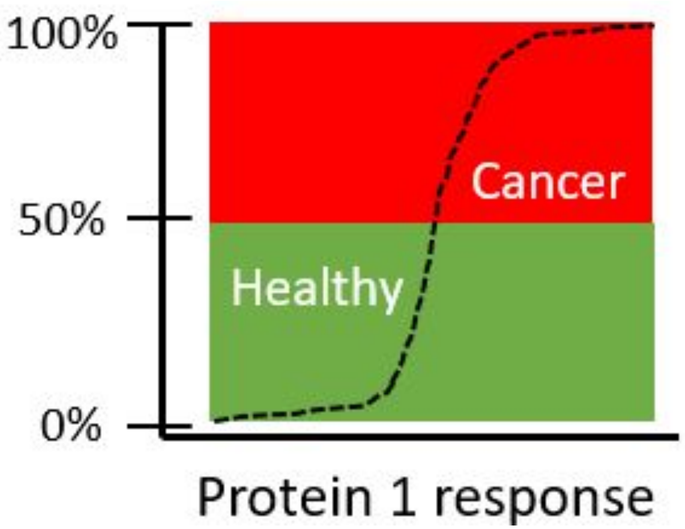
probability (that Y is in category k, given X

$$P(Y = k \mid X) = f(X)$$

output, response

category or class

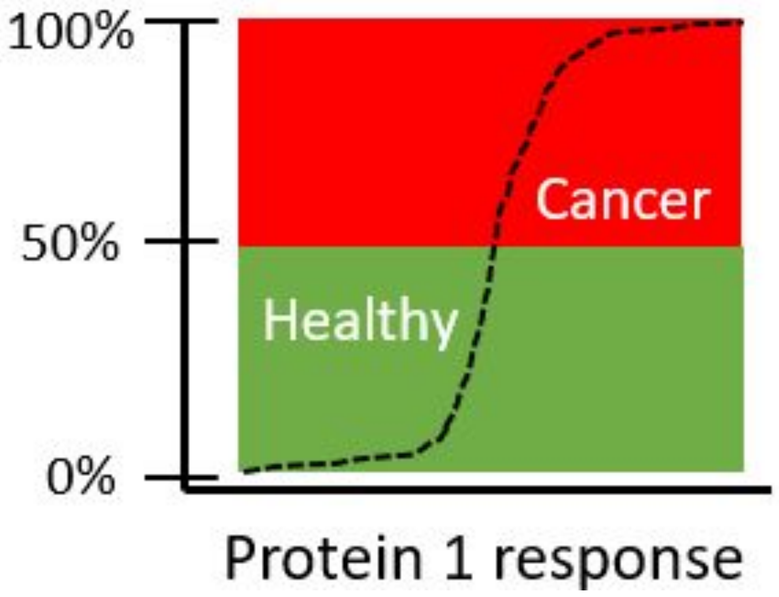


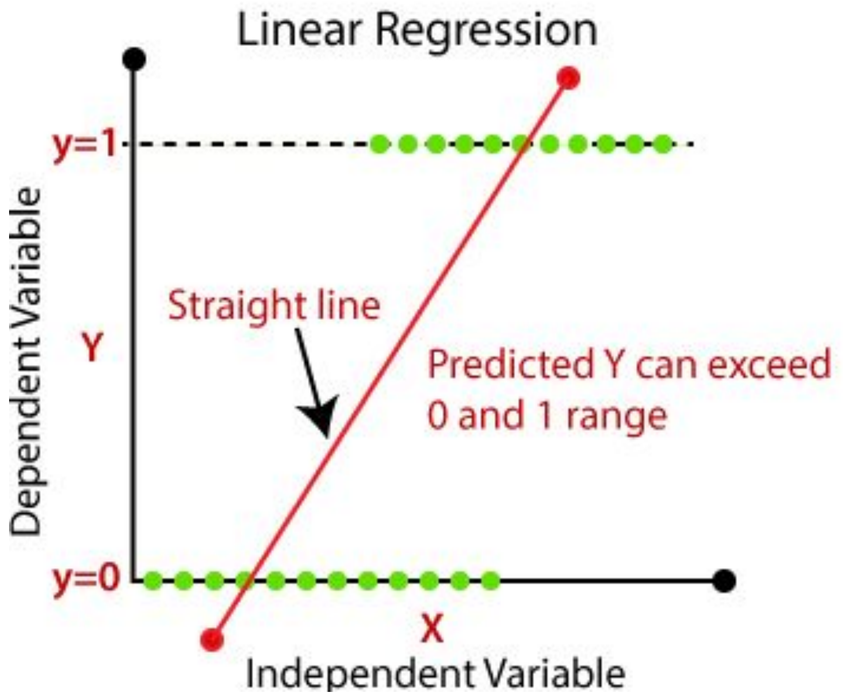


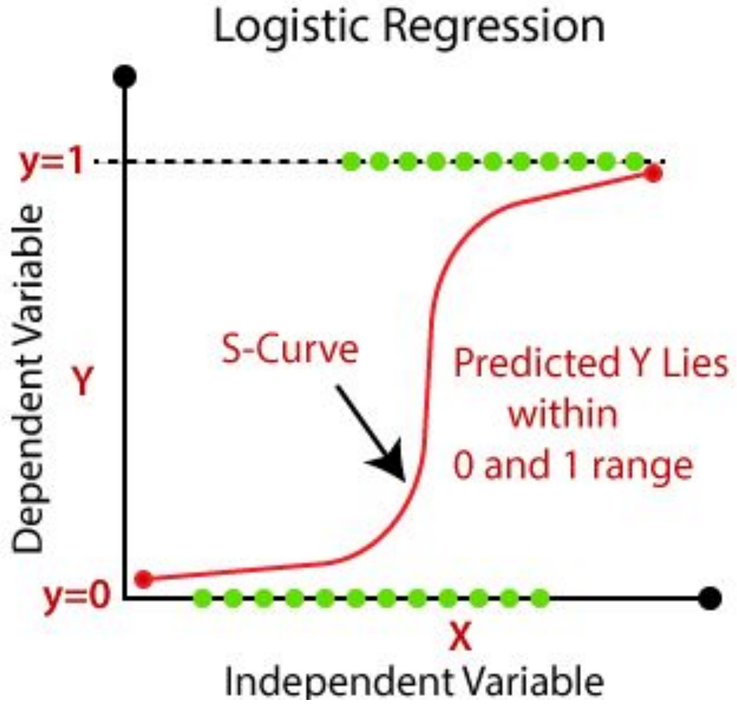
Logistic Regression

Logistic regression: models the relationship between predictors and the probability of an outcomes

- f is a *sigmoid function* (S-shaped curve) and outputs a probability
- Linear regression **CANNOT** accommodate **categorical** responses







Logistic Regression

<u>Simple</u> logistic regression model (one X), binary classification (outcome is 0 or 1)

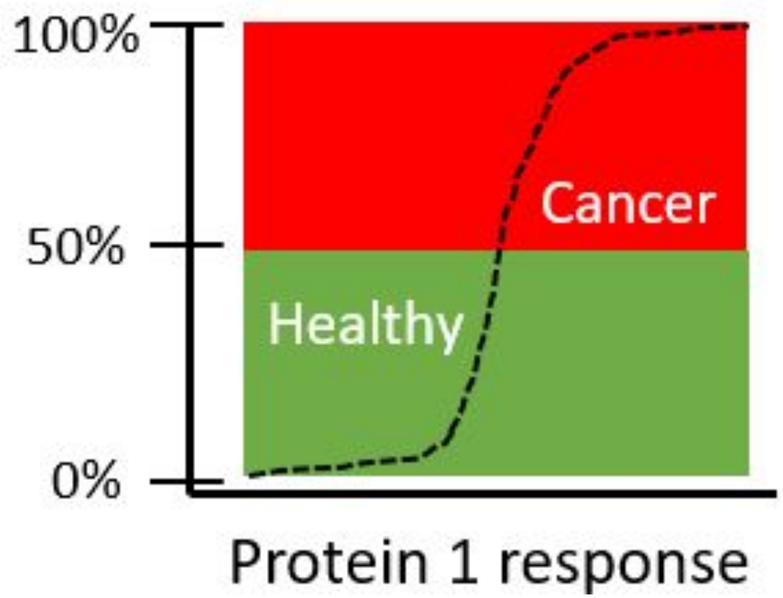
unknown *parameters*

probability (that Y is in category 1)

$$P(Y = 1 \mid X) = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

output, respons

predictors, features



Classification rule: $Y = \begin{cases} 1, & \text{if } P(Y = 1 \mid X) > 0.5 \\ 0, & \text{otherwise.} \end{cases}$

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course code:
WSTAPX

You want to predict whether or not you will get an A in COGS 9 this quarter. Which model should you use?

(A) Multiple linear regression

(B) Simple linear regression

(C) Polynomial regression

(D) Logistic regression

(E) Any of the above could work

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course code:
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You want to predict whether or not you will get an A in COGS 9 this quarter. Which model should you use?

(A) Multiple linear regression

(B) Simple linear regression

(C) Polynomial regression

(D) Logistic regression

(E) Any of the above could work

Supervised Learning Models

Regression: predicting quantitative responses

Models:

- Linear regression (simple & multiple)
- Non-linear regression
- Ridge & lasso regression

Both regression & classification:

- Decision Trees
- Random Forests & Boosting
- K-Nearest Neighbors
- Neural Networks

Classification: predicting qualitative (categorical) responses

Models:

- Logistic regression
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)
- Naive Bayes
- Support Vector Machine (SVM)

Takeaway: Your choice of model depends on the *outcome* you are predicting as well as the *relationship* between the input and output!